

USB Product ID / host tool settings

The USB bootloader documentation says the example uses USB-HID with Product ID 0x0000. The application hex file from the host PC: [Source: Minichip/Hermes USB HID bootloader](#).

You must match the firmware settings.

```
firmware:
    - > Basic
host tool:
    C
    c
```

The host tool is located at:

```
hwtool -J:\HidFirmware\*.jar
```

and the normalized application hex file to the USB HID bootloader: [Source: Hermes USB HID Host Application Bootloader](#).

Example:

```
usb product ID 23 match the Product ID entered into the Unified Host
```

9. Build, program, and validate

5. Build, program, and validate

The first time, the MCU is blank, so it uses MCU cannot bootstrap itself.

So the initial programming flow is:

1. Connect PXC14 to 4 ISP header.
2. Program the USB bootloader into the PIC20C4500B0M.
3. Merge the board while holding the bootloader button (jumper).
4. Connect USB to PC.
5. Open Microchip Unified Host tool.
6. Load the test_app_hex file.
7. Program over USB.
8. Repeat board.
9. Verify the application load blink.

Your validation checklist should include:

- Power LED turns on
- PXC14 can detect the device
- Bootloader/USB program successfully
- USB device enumerates on PC
- Unified Host tool detects program
- Unified Host blinks after reset
- Bootloader button/jumper forces bootloader mode
- Application cannot overwrite bootloader region

From <https://www.microchip.com/zh-hk/development-tools-kit/pic20c4500b0m> Bootloader: Device Name



THIS
ONC

[illegible][illegible]

Place the components illustrated in [Figure 2-2](#) within one quarter inch of each other from the SBC 3000.

FIGURE 2-2 EXAMPLE OF MCLR PIN CONNECTIONS



6/24/2025
From John T Lewis

The headers will be on the board and will have about 250mils of vertical clearance. Components can be placed in the area under which the socket will plug into the board. This will make it simple to place filter caps and the oscillators as close as possible to the corresponding pins. Do what you can with placement and routing and we can work through things together to finalize. It would be nice to place the PCB order by next Tuesday.

The parts arrived today from digikey.

A couple of quick tutorial resources for PCB layout and routing:

Design Rules:
https://youtu.be/TDkU3TmPE8Q?e=14_8WVVB3QFaf6

PCB Layout:
<https://youtu.be/gaf9/bN7ZcL8H0M?e=4VKJcC2Nds41vYU>

PCB Routing:
<https://youtu.be/X4e8XyUnW6P?e=dlH6TforSS08ra>

Design Rules – Altium overview
https://www.altium.com/documentation/altium-designer/pd/design-rule-types?e=14t4-AlmBQoc_y6D7VvDGEFgGmGqG5G8kzyNbuM_RfDVmGEPH2X0X2aFz

Questions:
Can we create a 3d printed enclosure with Luke and Lucas.

Great - we'll get together Monday and go over everything. There is another 8MHz crystal oscillator part that we can use for the board. It is a smaller part and nicer than the existing one. Easy to change out in the schematic at any time and then update the PCB layout. The capacitors for the oscillator will likely be the same values, you can check the PIC32MX55x data sheet for specifics. We can always change the values on the board if needed.

If you want me to look over the schematic and layout this weekend, let me know.

Crystal info:

Digikey Part Number: 110-407F39E008M0000CT-ND

Data sheet attached, note that the connection is for the crystal is at pins 1 and 3.

Part model info here: <https://www.digikey.com/en/models/263700074ab-muul>

Only feedback so far is to place the filtering capacitors as close as possible to the defined pins. Same goes for the oscillators and their filtering caps

Oh -- one more thing -- the Click Bus connector hosts boards that are typically no wider, but often longer. Might be good to orient that part perpendicular to the socket connectors.

A few thoughts:

- We'll stick with 4-layer, but try to keep routing on top and bottom. We'll try to keep layers 2 and 3 as ground plane. Keep the ground planes/copper pours on all layers about 40 mils from the edge of the board. It's explain why later.
- A note on Vias - Vias will need to go through the board, JLPCB does not support blind or buried vias.
- For power networks, sometimes works best to have a net that runs around the edge of the board -- also around 40 mils from edge.

Glossary

<https://cadcam.us/a/glossary>

- **Gerber files:** These files show where copper, solder mask, silkscreen, and solder paste are applied to the board during fabrication and assembly. These files are used to create stencils for fabrication.
- **PCB Panel drawing:** A PCB panel is usually created by the fabricator, although sometimes they provide templates that the designer can use to create a panel. This drawing shows how the individual boards will be laid out the panel and where they can be broken apart from each other.
- **Drill file:** This file lists the coordinates and hole sizes for all drill holes that will appear on the PCB.
- **Pick-and-place file:** This file lists the coordinates for all components on the board so that they can be automatically placed during assembly.
- **Fabrication and assembly drawings:** These drawings show the location of components, drill hits, and layer order in the PCB stackup.

- **Mechanical layers:** These store 2D projections of 3D models for components, allowing you to see component outlines that span beyond the limits of component pads.
- **Board outline layer:** This layer shows the routing path that will be used to cut out the board during manufacturing.
- **Overlay layers:** The top and bottom overlay layers store the silkscreen for components. You may remember that some silkscreen features were defined when you create components. However, you can also add custom silkscreen as text if needed in your design.
- **Assembly layers:** This set of layers is used to define paste mask and solder mask apertures in your PCB layout. You might need to modify information on these layers after you finish your PCB layout.

From: <https://education.altium.com/blogs/144-178/feature/1774030p>

- **Dielectric constant:** This will determine how signals propagate in your system and the level of losses seen by signals as they travel. In particular, it will play an important role in determining the impedance of your traces.
- **Glass transition temperature (Tg):** This is the temperature at which a PCB laminate begins expanding at a faster rate when it begins heating above a certain temperature. Low-end laminates will have Tg = 130 °C. High reliability laminates should have Tg = 150 °C or 170 °C.

From: <https://education.altium.com/blogs/144-178/feature/1774030p>

Tricks / Hotkeys:

Net Highlighting

One way to see the full extent of the connections made by a trace on multiple layers is to hover over a trace, pad, polygon, via, or other object, and hold down the Shift key. While the Shift key is depressed, every object that is part of the net will appear with a white outline.

From: <https://education.altium.com/blogs/144-178/feature/1774030p>

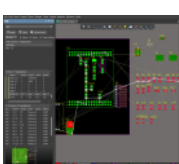
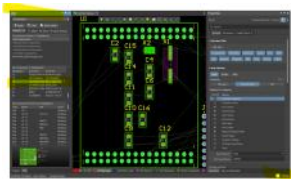
In order to choose which objects will be selected as you navigate a PCB layout, you'll need to use the selection filter. This feature can be found in the Properties Panel.

Whenever you need to deselect an object, you can Shift + Click on the selected object, or you can click on blank space (the grey background in the PCB) to deselect everything.

From: <https://education.altium.com/blogs/144-178/feature/1774030p>



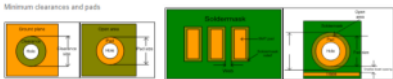
Finding / selecting tools
PCB panel



Manufacturing

We're using:
JLPCB

Manufacturing capabilities example / Terminology



Vid on pcb manufacturing:
[Manufacturing Tour with Q&A - Wirth Electronics - AltiumLive 2020](https://www.youtube.com/watch?v=Q8A_Wy0mEhE&list=PL500)



IPC-4101 standards and slash sheets

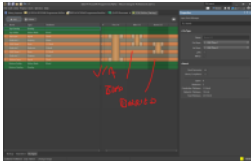
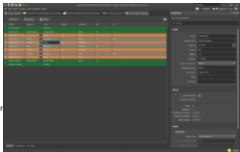
IPC-4101 standards and slash sheets	
Drillings (Holes), Drill Diameter and Depth	Pre-Drillings (Holes) from 0.015" (PC-4101) before December 31, 2014
Current Revision Specifications Sheet	
10	10A, 10B, 10C, 10D, 10E, 10F, 10G, 10H, 10I, 10J, 10K, 10L, 10M, 10N, 10O, 10P, 10Q, 10R, 10S, 10T, 10U, 10V, 10W, 10X, 10Y, 10Z
20	20A, 20B, 20C, 20D, 20E, 20F, 20G, 20H, 20I, 20J, 20K, 20L, 20M, 20N, 20O, 20P, 20Q, 20R, 20S, 20T, 20U, 20V, 20W, 20X, 20Y, 20Z
30	30A, 30B, 30C, 30D, 30E, 30F, 30G, 30H, 30I, 30J, 30K, 30L, 30M, 30N, 30O, 30P, 30Q, 30R, 30S, 30T, 30U, 30V, 30W, 30X, 30Y, 30Z
40	40A, 40B, 40C, 40D, 40E, 40F, 40G, 40H, 40I, 40J, 40K, 40L, 40M, 40N, 40O, 40P, 40Q, 40R, 40S, 40T, 40U, 40V, 40W, 40X, 40Y, 40Z
50	50A, 50B, 50C, 50D, 50E, 50F, 50G, 50H, 50I, 50J, 50K, 50L, 50M, 50N, 50O, 50P, 50Q, 50R, 50S, 50T, 50U, 50V, 50W, 50X, 50Y, 50Z
60	60A, 60B, 60C, 60D, 60E, 60F, 60G, 60H, 60I, 60J, 60K, 60L, 60M, 60N, 60O, 60P, 60Q, 60R, 60S, 60T, 60U, 60V, 60W, 60X, 60Y, 60Z
70	70A, 70B, 70C, 70D, 70E, 70F, 70G, 70H, 70I, 70J, 70K, 70L, 70M, 70N, 70O, 70P, 70Q, 70R, 70S, 70T, 70U, 70V, 70W, 70X, 70Y, 70Z
80	80A, 80B, 80C, 80D, 80E, 80F, 80G, 80H, 80I, 80J, 80K, 80L, 80M, 80N, 80O, 80P, 80Q, 80R, 80S, 80T, 80U, 80V, 80W, 80X, 80Y, 80Z
90	90A, 90B, 90C, 90D, 90E, 90F, 90G, 90H, 90I, 90J, 90K, 90L, 90M, 90N, 90O, 90P, 90Q, 90R, 90S, 90T, 90U, 90V, 90W, 90X, 90Y, 90Z
100	100A, 100B, 100C, 100D, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L, 100M, 100N, 100O, 100P, 100Q, 100R, 100S, 100T, 100U, 100V, 100W, 100X, 100Y, 100Z

AGC SLASH SHEET REFERENCE TABLE	
SLASH	SLASH
10A	10B
10C	10D
10E	10F
10G	10H
10I	10J
10K	10L
10M	10N
10O	10P
10Q	10R
10S	10T
10U	10V
10W	10X
10Y	10Z



opening your PCB file, and then select the Design → Layer Stack Manager

- Set plane and signal layers
- Specify core and prepreg layers
- Define via transitions
- Calculate impedance
- Specify materials constants
- Select copper weight on each layer



Ground Plane Layer

A ground plane layer, or simply a ground plane, is just like its name suggests: It acts like a ground connection for your circuits and components, meaning it sits at 0 V reference in the PCB. It provides other benefits as well, and the presence of at least one ground plane is required in many designs.

Some of the reasons a ground plane is used include:

- It provides shielding against external noise originating from radio frequency sources
- It provides a path for heat transport away from hot components
- When used with an adjacent power plane, the pair of planes will act like a big capacitor that helps ensure stable DC voltage
- It ensures the inductance in your interconnects will be low enough that the design will not generate or receive excessive EMI
- It helps ensure the impedance you calculate for your traces is the same everywhere in your board

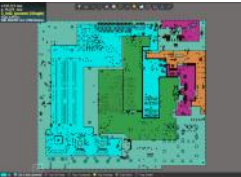
The last point is probably the most important as it is critical for routing common computer interfaces (e.g., USB, HDMI, Ethernet, etc.). When working with these high speed digital protocols, or with high frequency signals used in wireless systems, a ground plane is required to ensure signal integrity during routing. We'll discuss this more in a later unit; just know that you will need to include at least one dedicated ground plane when using these protocols.

From <https://education.abl.com/courses/5624786/lectures/72227958>

Power Plane Layer

Power layers are intended to provide DC or AC power to components in your board. A power plane can be connected to a positive or negative voltage output from your power supply, depending on the needs of your design. Power planes are normally used with a ground plane on an adjacent layer so that they can provide stable power by acting like a big capacitor.

One common practice is to split a power plane into different sections, which can supply different voltages to different parts of the PCB. If you plan to do this with your power polygons, then you need to define that layer as a signal layer in your PCB stackup. An example from the MiniPC project is shown below.



These colored regions in Layer 3 in the MiniPC project are all different power regions. Each of these carries a different voltage/current to different sets of components in the PCB layout.

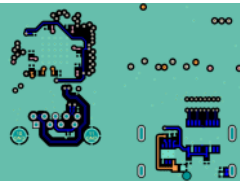
For high speed designs, such as in the MiniPC project, it's best to put a uniform power plane that sits adjacent to a uniform ground plane. In the MiniPC project, there is a large power plane on Layer 6, as well as smaller polygons used to supply power in Layer 3. In both cases, there is a uniform ground plane on the adjacent layer.

From <https://education.abl.com/courses/5624786/lectures/72227958>

Signal Layers

Every layer that isn't dedicated to a power plane or ground plane is a signal layer. Inside your ECAD software, you'll only be able to route traces on layers that are designated as signal layers. When there are multiple signal layers in your PCB, you'll need to define vias between layers so that traces can be routed between different layers.

Just because a layer was designated as a signal layer doesn't mean that the layer can't have ground or power regions. If you want to place power or ground on a signal layer, you need to place a copper polygon on the layer. Your design software will automatically define this as a region of copper, and you can connect the region to a power, ground, or signal net during routing. Another option that is available is to place sections of polygon, each connected to different power and ground nets without any traces. An example is shown below.



From <https://education.abl.com/courses/5624786/lectures/72227958>

copper-to-copper spacing: 4 mils
via hole-to-via hole spacing: 8 mils
solder mask sliver: 5 mils



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Stand-alone courses



Unit 3 - PCB Layout

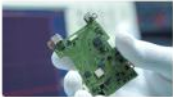
100% COMPLETE



Unit 2 - Schematics and Libraries

✓ Complete

Revisit course →



Unit 1 - Introduction to PCB Design

✓ Complete

Revisit course →

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